Version Control

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1. What is Version Control?

Version control is a system that tracks changes to files, enabling collaboration and the ability to revert to previous versions if necessary. It is essential in software development, ensuring that every modification is logged and multiple people can work on the same project without conflicts.

2. Why is Version Control Important?

Collaboration: Multiple developers can work simultaneously without overwriting each other's changes.

Backup: All changes are stored, allowing recovery of earlier versions.

Traceability: Tracks who made changes and why.

Branching and Merging: Allows feature development and testing without affecting the main project.

Reduced Risk: Protects against data loss and corruption.

3. Types of Version Control

a. Local Version Control

Tracks changes locally on a developer's machine. Example: RCS.

b. Centralized Version Control (CVCS)

Stores files on a central server. All clients share one repository. Examples: CVS, SVN.

c. Distributed Version Control (DVCS)

Every developer has a complete copy of the repository, allowing independent work. Example: Git.

4. What is Git?

Git is a distributed version control system widely used for managing code in projects. It allows multiple developers to work on the same codebase, track changes, and manage branching and merging efficiently.

Distributed: Every developer has a full copy of the repository.

Branching: Enables creation of isolated work areas.

Speed: Local operations are fast.

Data Integrity: Ensures data accuracy using checksums.

5. Important Git Commands

git init: Initializes a new Git repository.

git clone: Copies a remote repository to the local machine.

git add: Stages changes for the next commit.

git commit: Saves a snapshot of staged changes.

git status: Shows the current state of the working directory.

git push: Uploads local changes to a remote repository.

git pull: Fetches and merges changes from a remote repository.

git branch: Lists, creates, or deletes branches.

git checkout: Switches between branches.

git merge: Merges changes from one branch into another.

git log: Shows a log of commits.

git reset: Reverts changes or commits.